



What have these got
in common?.....



Challenge: Can you list any other examples?



Ecosystems





They are all.....
ECOSYSTEMS



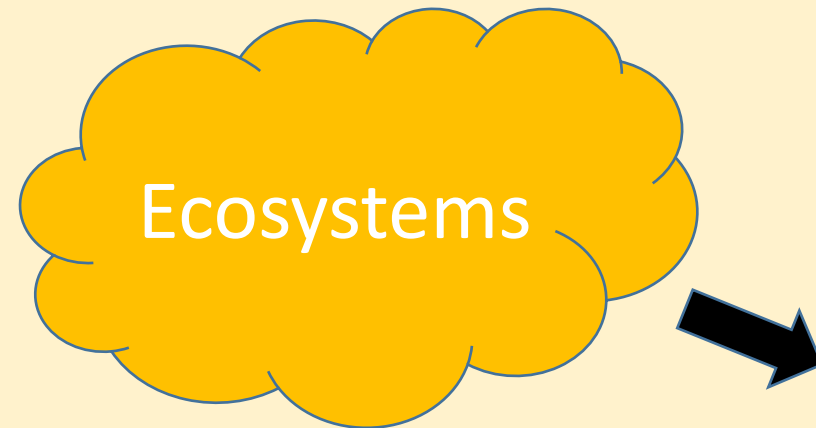
Title: Ecosystems

Starter: In pairs, draw a mind map in your book and write everything you already know about ecosystems.

Come up with your definition of what an ecosystem is.

In Singapore one museum showed a round glass bowl of water containing sand, and weeds and fish with light shining on it.

Was this an ecosystem?



They can be little as a pond or as a big as a rainforest.

Success criteria

Lesson Title	Working towards	Working at	Working beyond
What is an Ecosystem	I can name some living (biotic) and non-living (abiotic) parts of an ecosystem.	I can define key terms like community, population, habitat and explain how energy moves through a food web	I can explain the role of decomposers and interpret pyramids of numbers to show energy transfer in ecosystems.
Interactions within Ecosystems	I can identify some habitats found in the Sonoran Desert and other ecosystems.	I can describe interactions between organisms and their environment in different ecosystems.	I can explain the differences between habitats and ecosystems and how organisms are adapted to survive in these settings.
Adaptations and Competition	I can give examples of how plants or animals are adapted to their environments.	I can describe how adaptations help species survive and explain competition between organisms.	I can analyse mutualistic relationships and present research on how different species interact in ecosystems.
Invasive Species	I can name an invasive species and describe what a native species is.	I can describe how invasive species affect native plants and animals in an ecosystem.	I can explain why controlling invasive species is important and evaluate different methods of control.
Case Study - New Zealand (Debate)	I can identify an invasive species problem in New Zealand.	I can describe the ecological impact of invasive species and outline arguments for control.	I can construct and defend an argument in a debate using evidence about species control and conservation.
Bioaccumulation	I can state what an insecticide is and give a simple example.	I can describe how chemicals like DDT build up in organisms and impact the food chain.	I can explain why organisms at the top of the food chain have higher toxin levels and the environmental consequences.
Weather and Climate	I can describe the difference between weather and climate.	I can explain how weather affects plants and animals in an ecosystem.	I can analyse weather data to predict effects on local ecosystems and seasonal changes.
Climate and Atmosphere	I can name key gases in the atmosphere.	I can explain how greenhouse gases affect Earth's climate and describe human impacts.	I can evaluate how climate change may affect ecosystems and propose possible solutions.



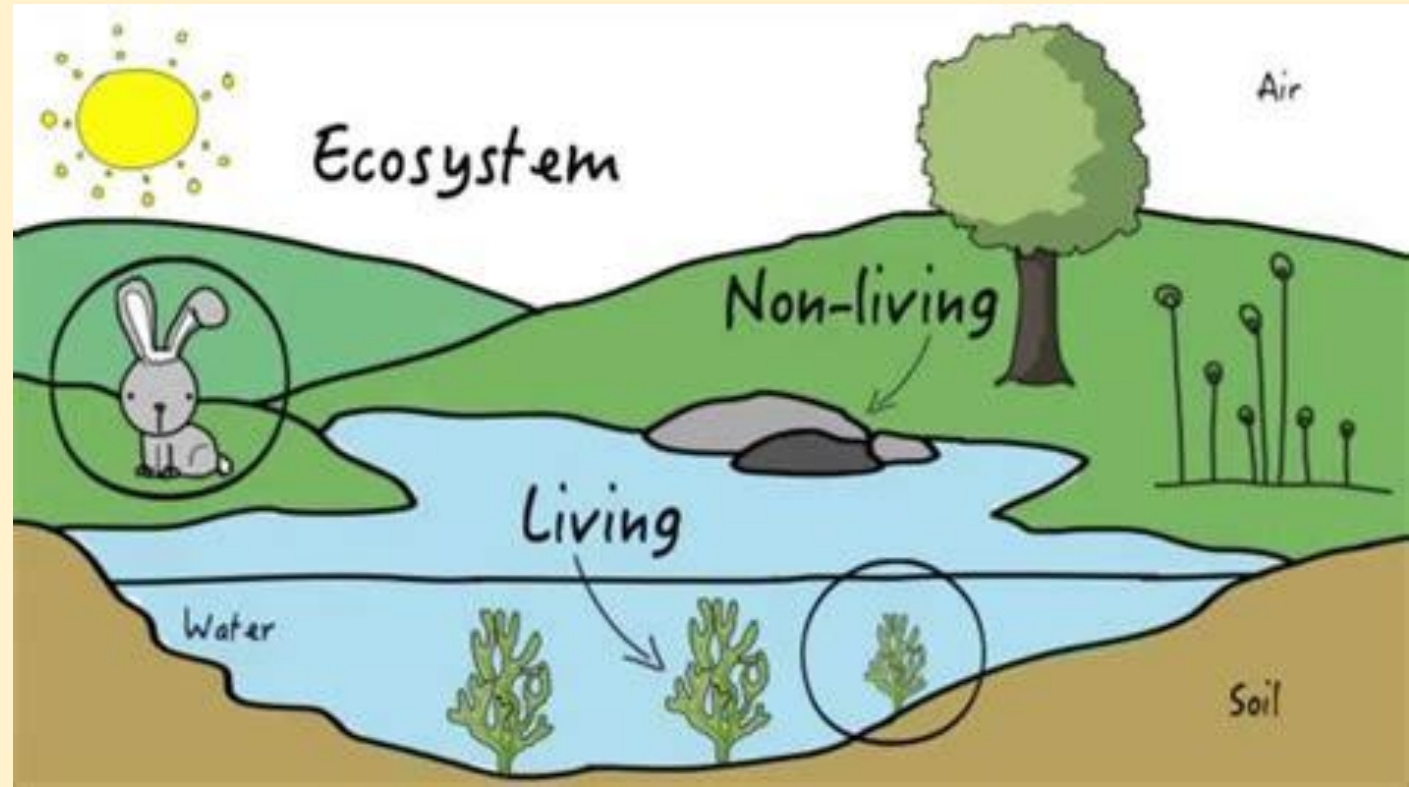
NAT GEO
WILD



What is an ecosystem?

In pairs, discuss for **2 minutes** what an ecosystem is and come up with your own definition.

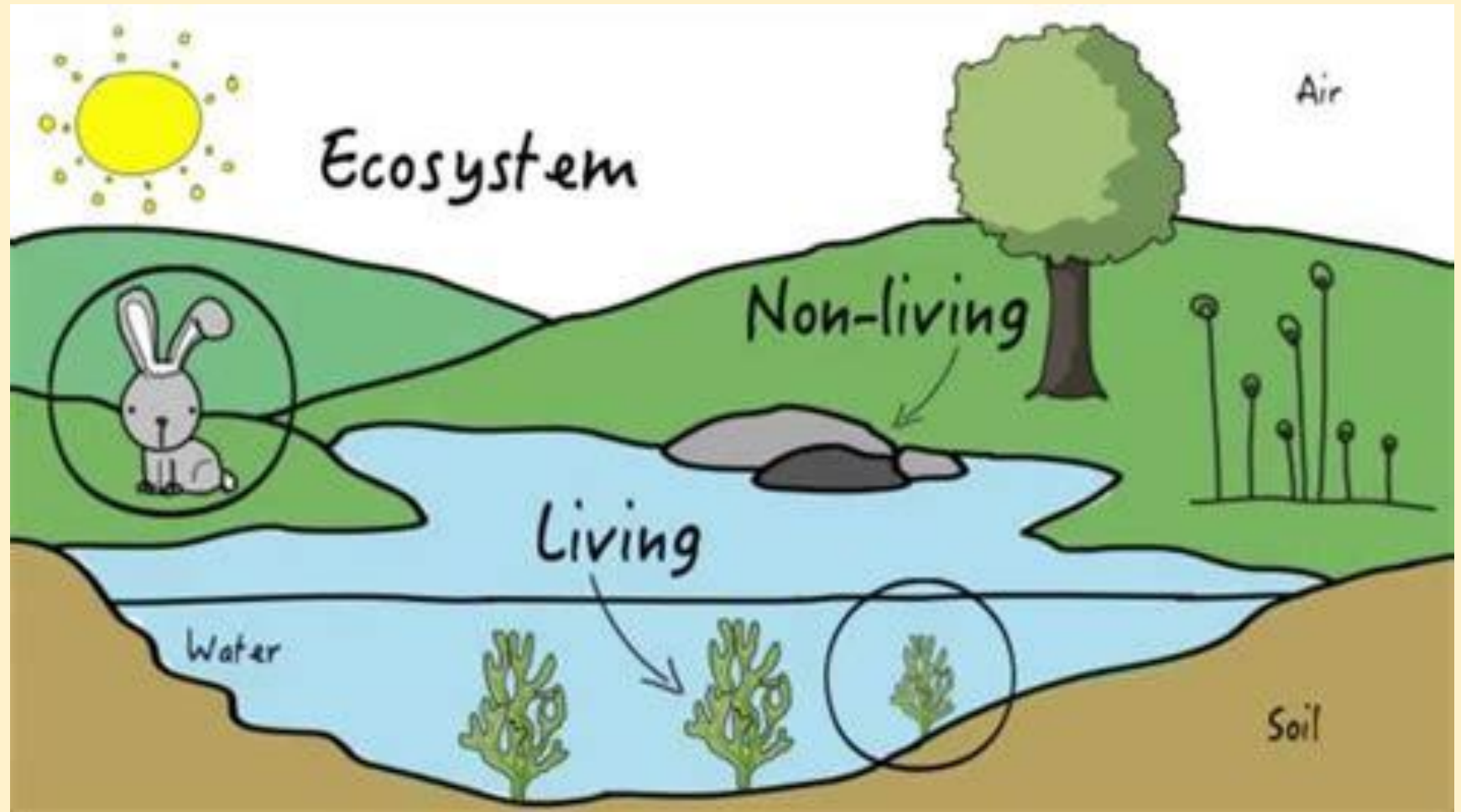
Be prepared to share your definitions!



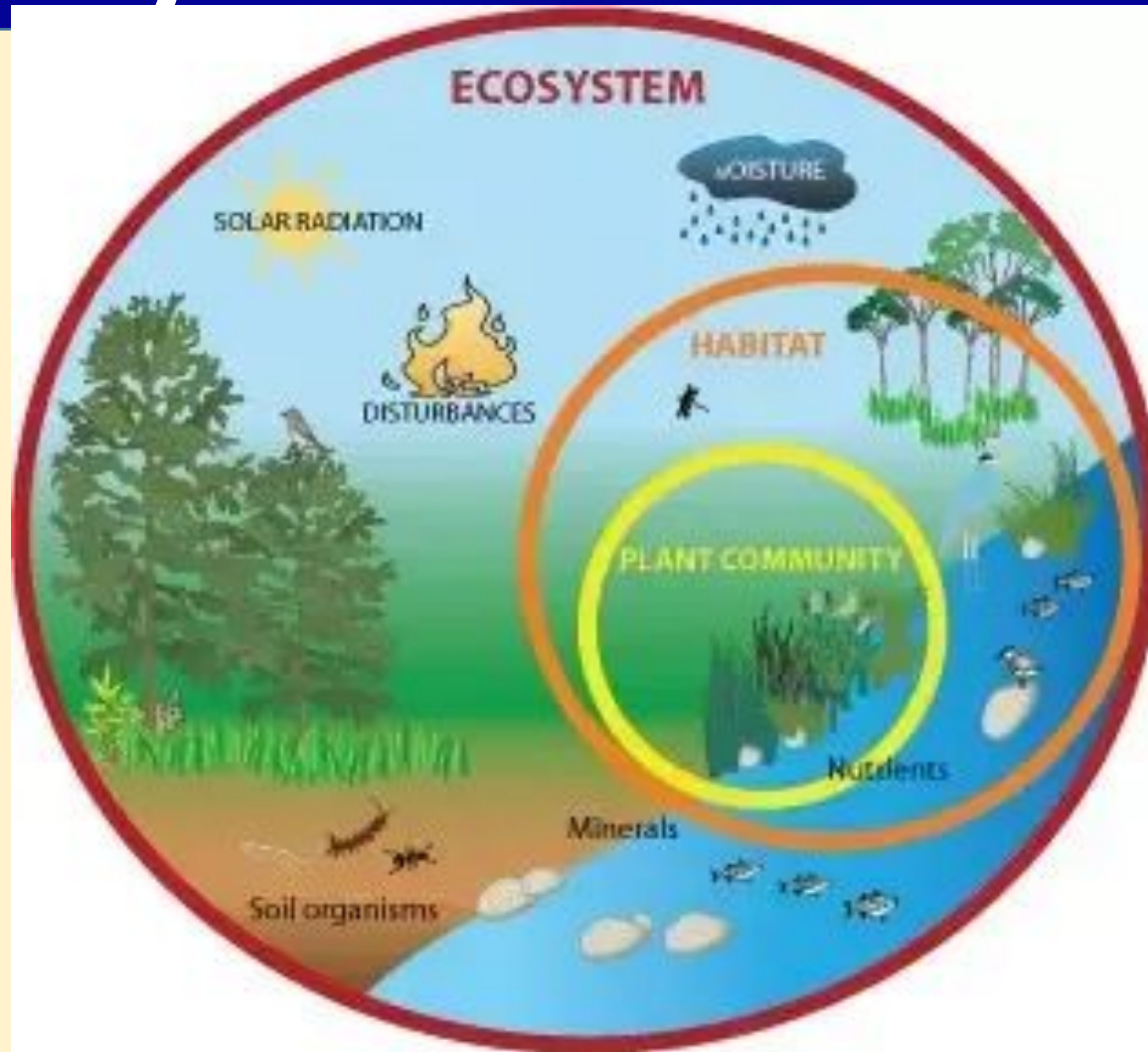
Ecosystem

A network containing all of the organisms (living - biotic) and their environment (non living - abiotic), interacting together, in a given area

e.g. a lake.



What is the difference between an ecosystem and a habitat?



So, what is the difference between an ecosystem and a habitat?

A habitat is

An ecosystem is



In Singapore (many years ago!) one museum showed a round glass bowl of water containing sand, and weeds and fish with light shining on it. Was this an ecosystem? **Yes**

So, what is the difference between an ecosystem and a habitat?

A habitat is the natural home or environment of an animal, plant, or other organism.

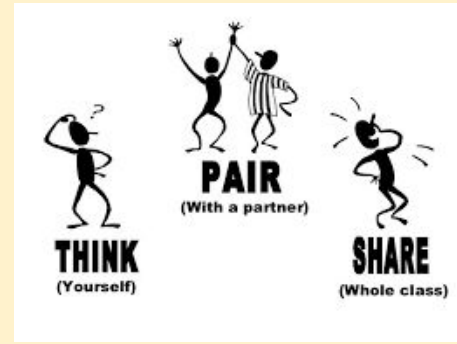
An ecosystem is a network of living organisms interacting with each other and their non-living environment.

Ecosystems contain biotic (living), parts, as well as abiotic (non-living) factors.

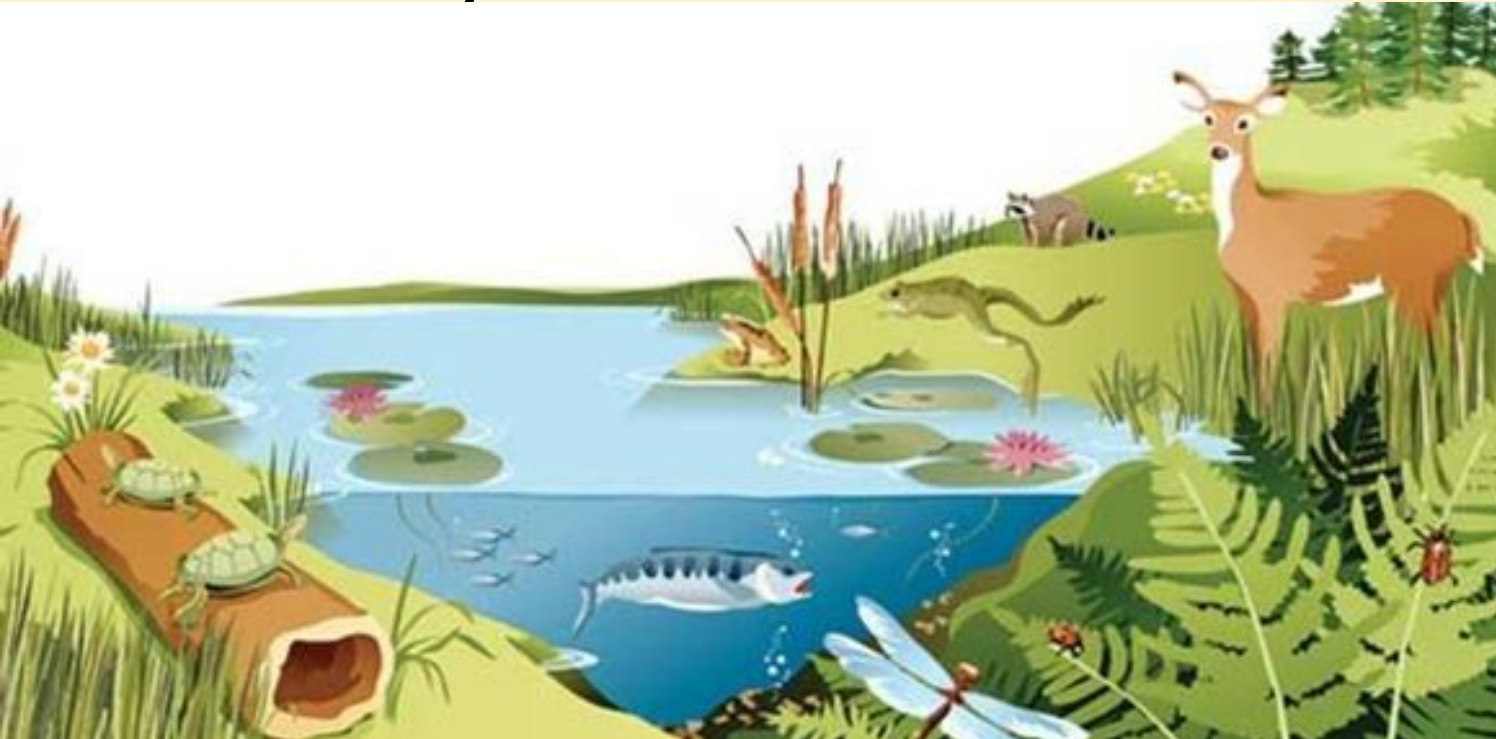
So

Ecosystem = Habitat(s) PLUS the organisms (living) PLUS the abiotic (non living) factors there

ECOSYSTEMS CONTINUED



- The organisms in an ecosystem are known as the **community**
- The area they live is called the **habitat**



**How many
animal types in
this community?**

**What's the
fishes habitat?**

7 species in this community. Fishes habitat is the pond.

Biotic and Abiotic

An ecosystem is made up of living (**biotic**) and nonliving (**abiotic**) things interacting with each other.



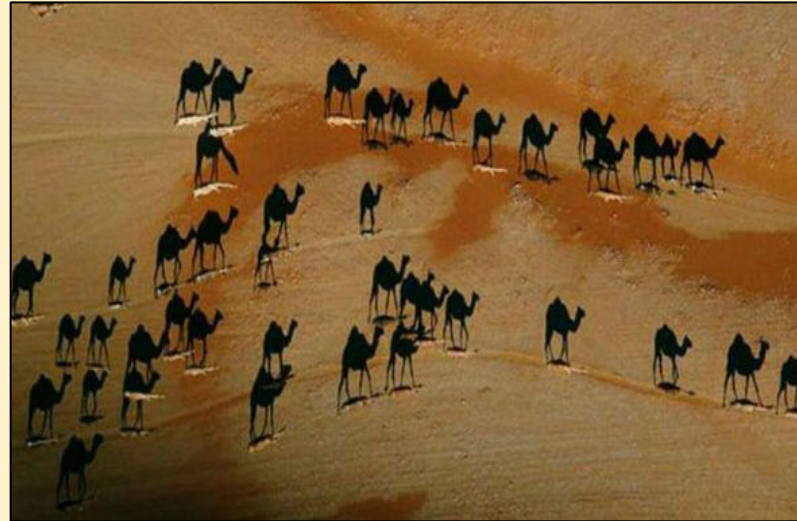
What are the
biotic and
abiotic
features here?

Abiotic (Non Living) Factors

What are examples of abiotic factors?

- Temperature
- Humidity / amount of water
- pH
- Light level / hours of light
-?

Name as many ecosystems as you can



Name as many ecosystems as you can



Tropical rainforest



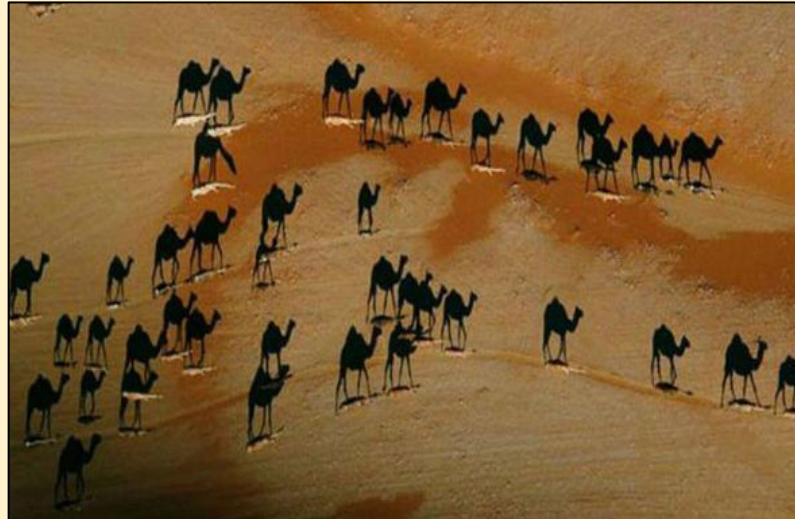
Rice Paddy



Sea ice in the artic



Savannah grassland



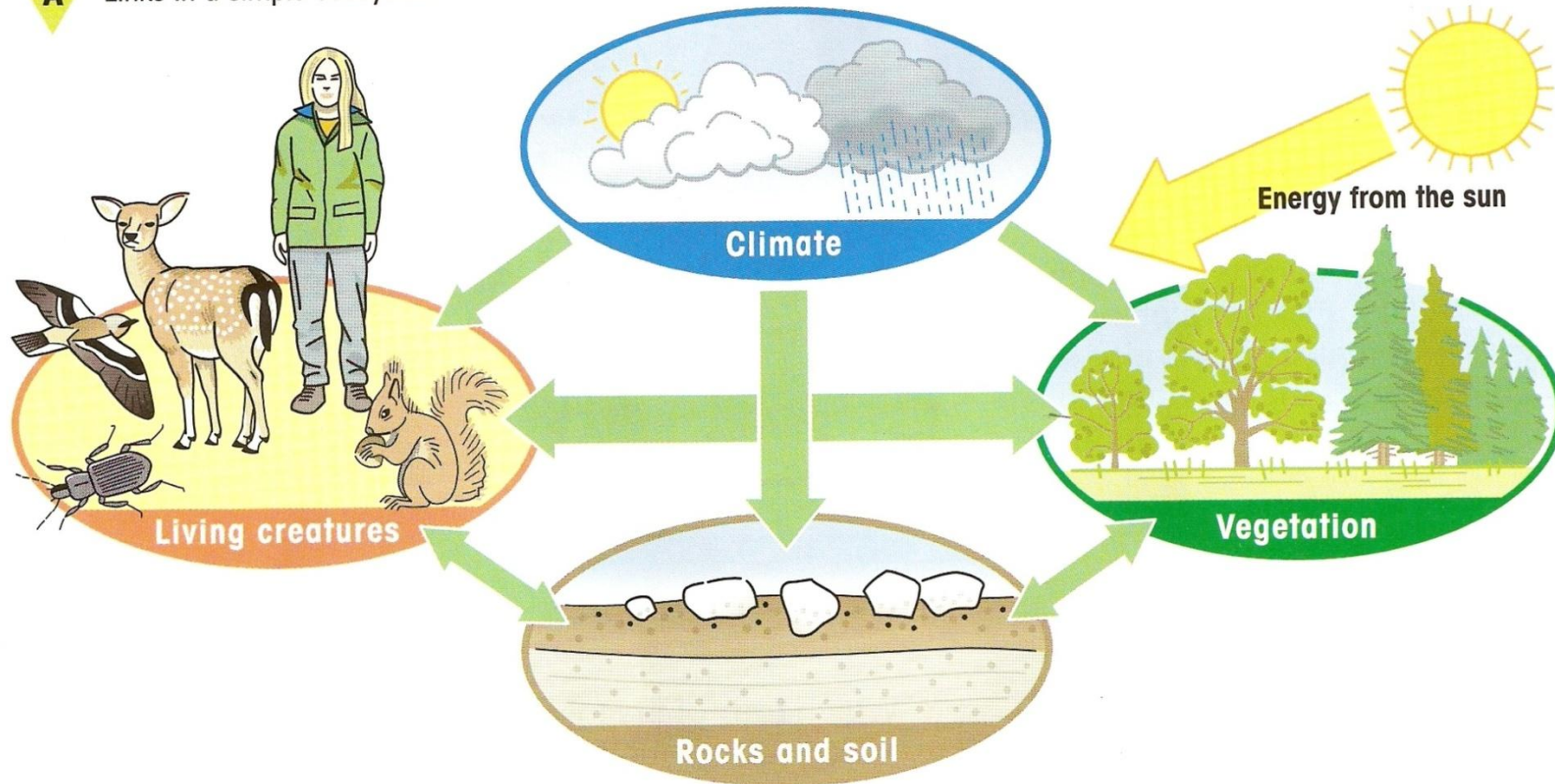
Desert



Mangrove Forest

What is an ecosystem?

A Links in a simple ecosystem



Activity: Explain
2 or 3 links
within an
ecosystem.

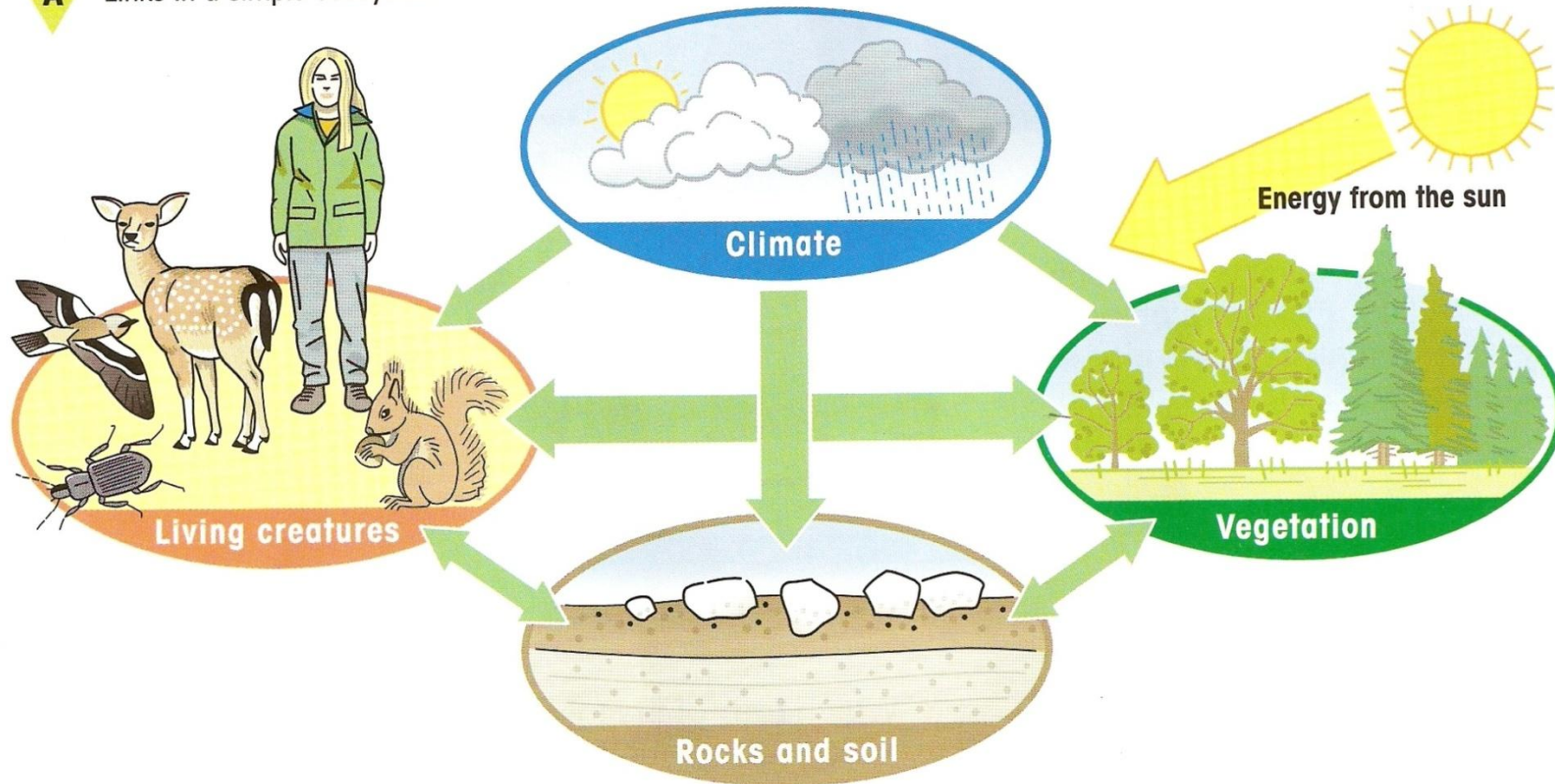
Write them in
your books.

Example: Animals are linked to the soil because

.....

What is an ecosystem?

A Links in a simple ecosystem



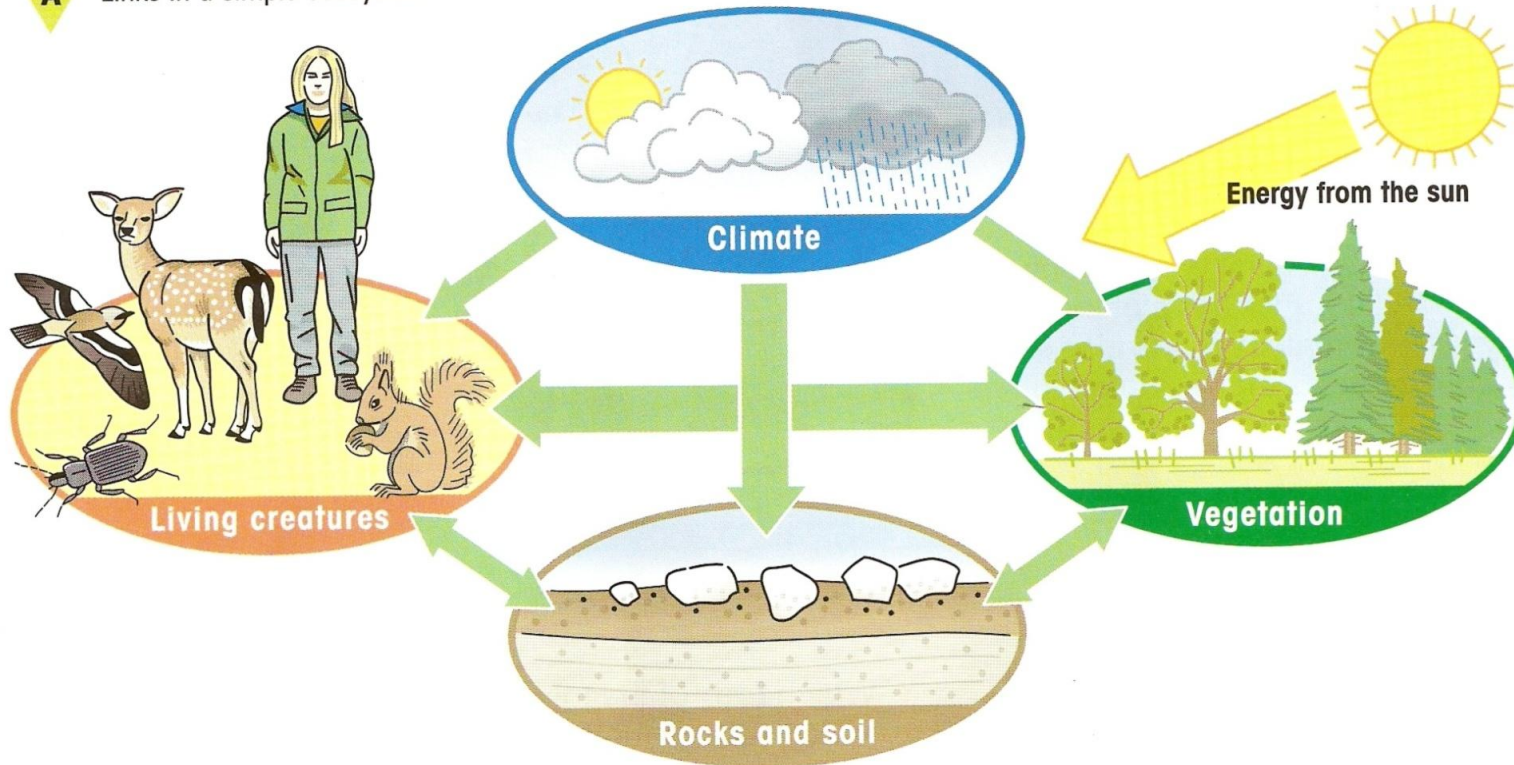
Activity: Explain 2 or 3 links within an ecosystem.

Write them in your books.

Example: Animals are linked to the soil because they may use it to provide shelter

What is an ecosystem?

A Links in a simple ecosystem



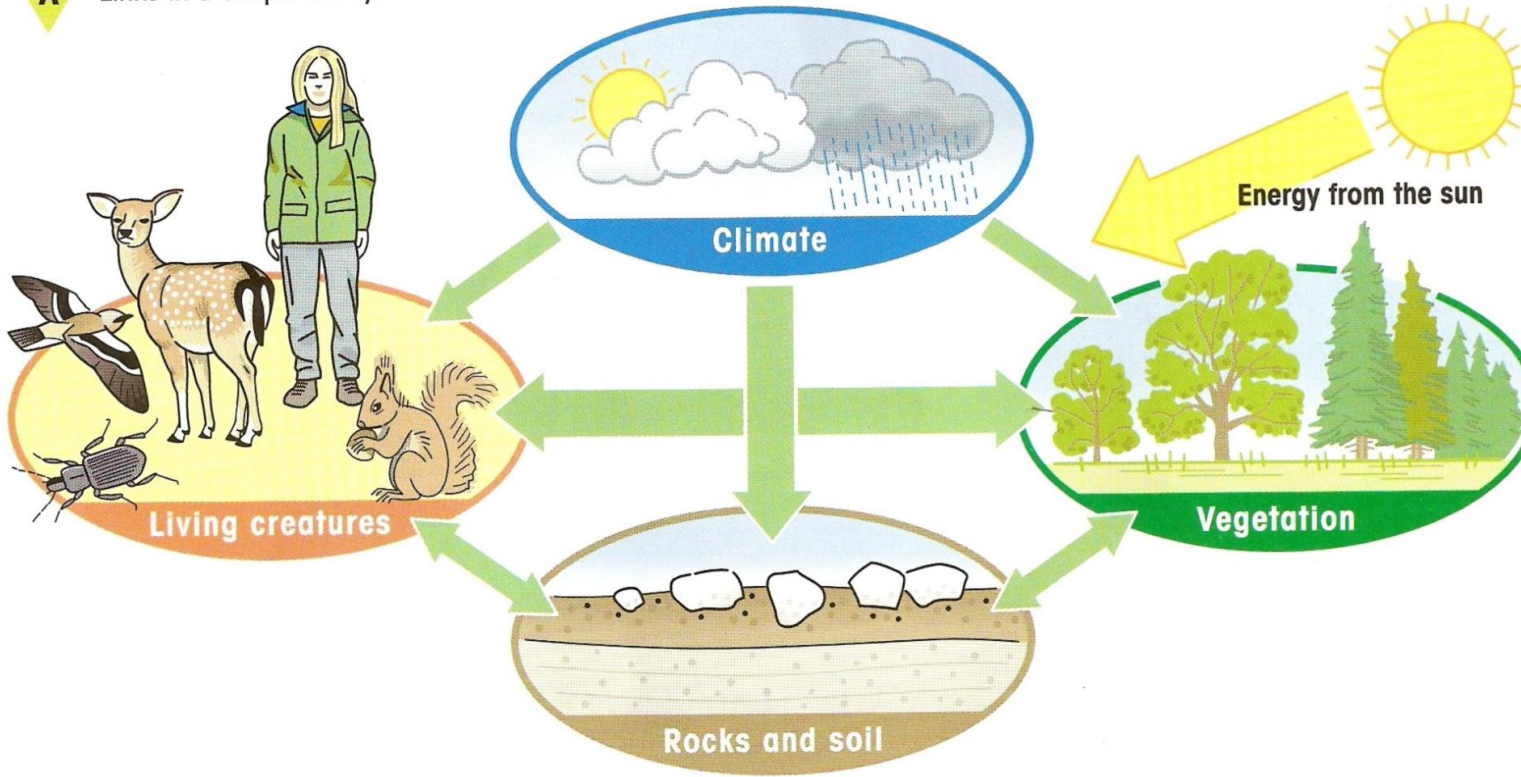
Your Turn

- 1 Plants are linked to the sun because ...
- 2 Climate is linked to vegetation because...
- 3 Living creatures are linked to vegetation because...
- 4 Soil is linked to living creatures because ...

8:00

What is an ecosystem?

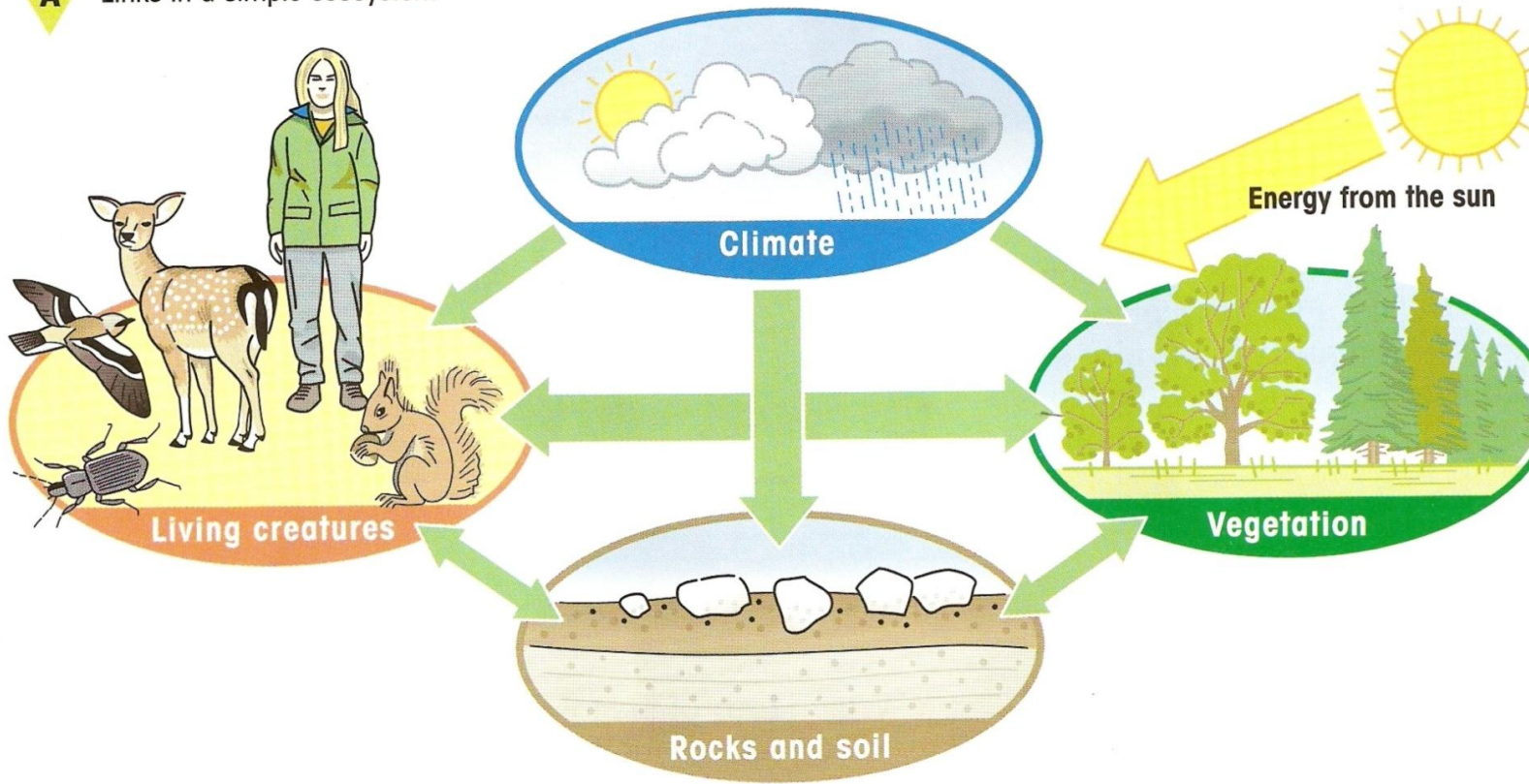
A Links in a simple ecosystem



1. **Plants are linked to the sun** because they use sunlight to make food through photosynthesis.
2. **Climate is linked to vegetation** because temperature and rainfall affect how well plants grow.

What is an ecosystem?

A Links in a simple ecosystem



3. Living creatures are linked to vegetation because animals eat plants for food and use them for shelter.

4. Soil is linked to living creatures because decomposers like worms and beetles break down dead plants and animals to return nutrients to the soil.

How are all the organism in an ecosystem linked together?

Simple answer: **FOOD!**

....who eats who!

... and how do we show that?

Food chains

A food chain shows the different organisms that live in a habitat, and what eats what.



Grasshopper



Snake



Eagle



Grass



Frog

Which way should the arrow point?



Activity: In your books draw this food chain in the correct order.

Challenge: Add another food chain for a different habitat.

Tip:- the arrow points to the organism that is doing the eating.

Food chains



Grass



Grasshopper



Frog



Snake



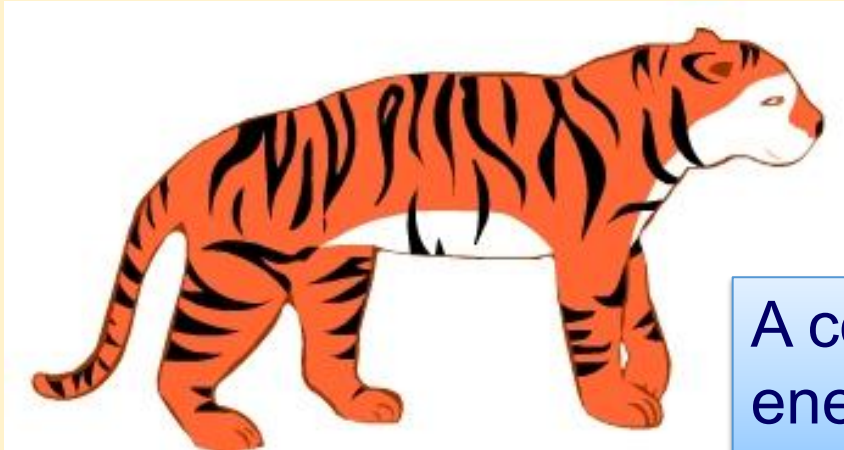
Eagle

What can we call the different parts of a food chain?

What must EVERY food chain contain?

Feeding types

A producer is an organism that makes its own organic nutrients, usually using energy from sunlight, through photosynthesis



A consumer is an organism that gets its energy by feeding on other organisms.

Food chains



Grass



Grasshopper



Frog



Snake



Eagle

Can you identify the producers and consumers in the food chain?

Food Chain



Grass



Grasshopper



Frog



Snake



Eagle

Secondary consumer

Tertiary consumer

Top predator

Producer

**Primary
consumer**

Activity: Place the key words onto your food chain.

2 MINUTES

Food Chain



Grass



Grasshopper



Frog



Snake



Eagle

Producer

**Primary
consumer**

**Secondary
consumer**

**Tertiary
consumer**

**Top
predator**

Food Chain Vocabulary

What is a Producer?

Eg. any plant

What is a Consumer?

Eg. any animal

How could we group different consumers?

Consumers

Consumers can be grouped into different types:

Herbivores

An animal that gets its energy by eating plants.



Carnivores

An animal that gets its energy by eating other animals.



Decomposers

An organism that gets its energy from dead or waste organic matter.



Consumers

You may have also said:

- Prey
- Predator

An animal could be both predator and prey
(eg a frog eats, but is eaten by).

A TOP predator is just a predator at the TOP of the chain - eg

Quick Check

1. Can you create **food chains**? Arrange these organisms into 2 food chains:

Challenge: *Label what we call the various organisms in the chain*

A

Grasshopper

Hawk

Snake

Flower

Rat

B

Mackerel

Whale

Phytoplankton (a plant)

Sea bass

Zooplankton

2. What does the arrow mean?

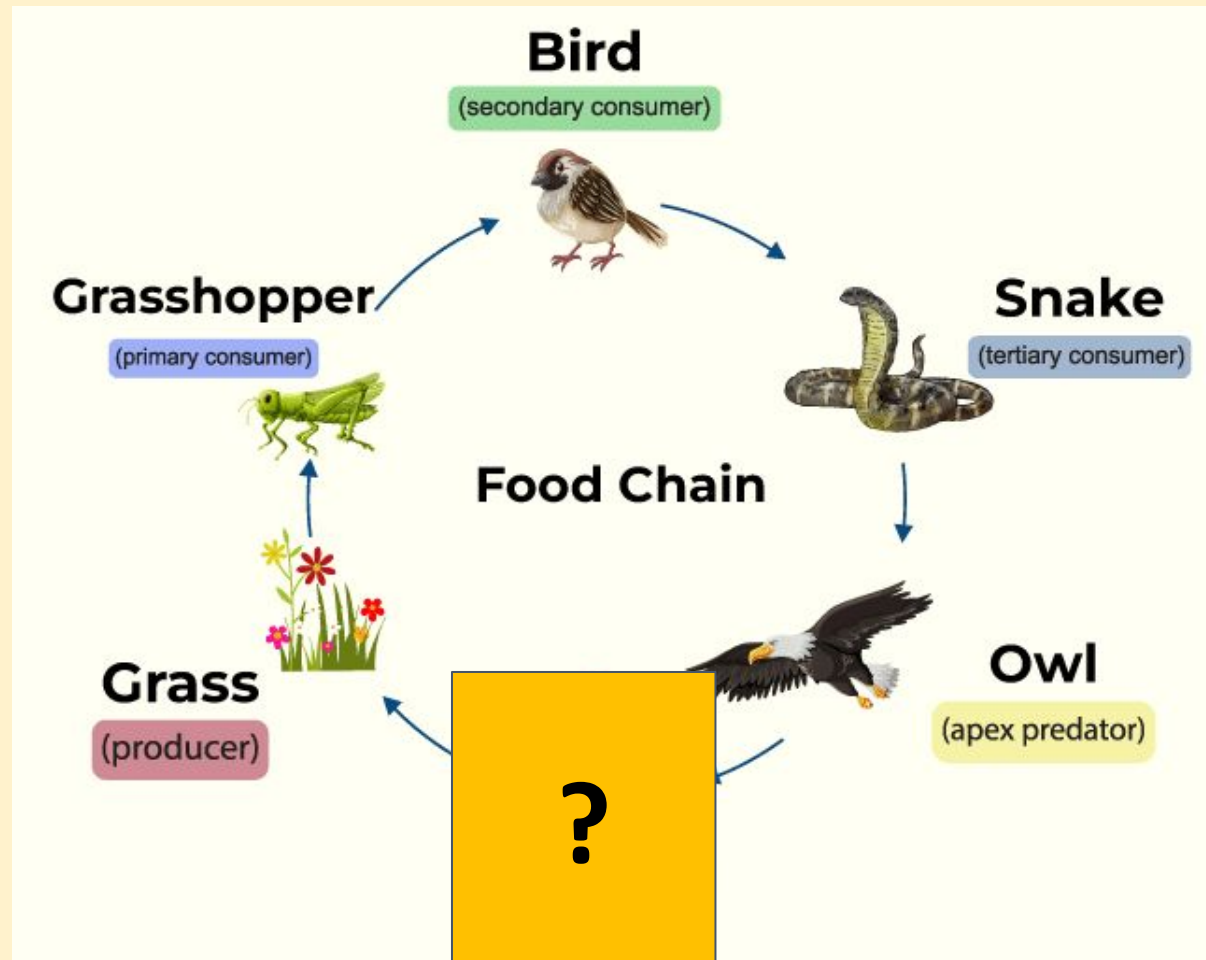
The arrow shows us the direction the energy is being transferred

3. So which way does the arrow always go?

It goes to the animal doing the eating/ towards top predator

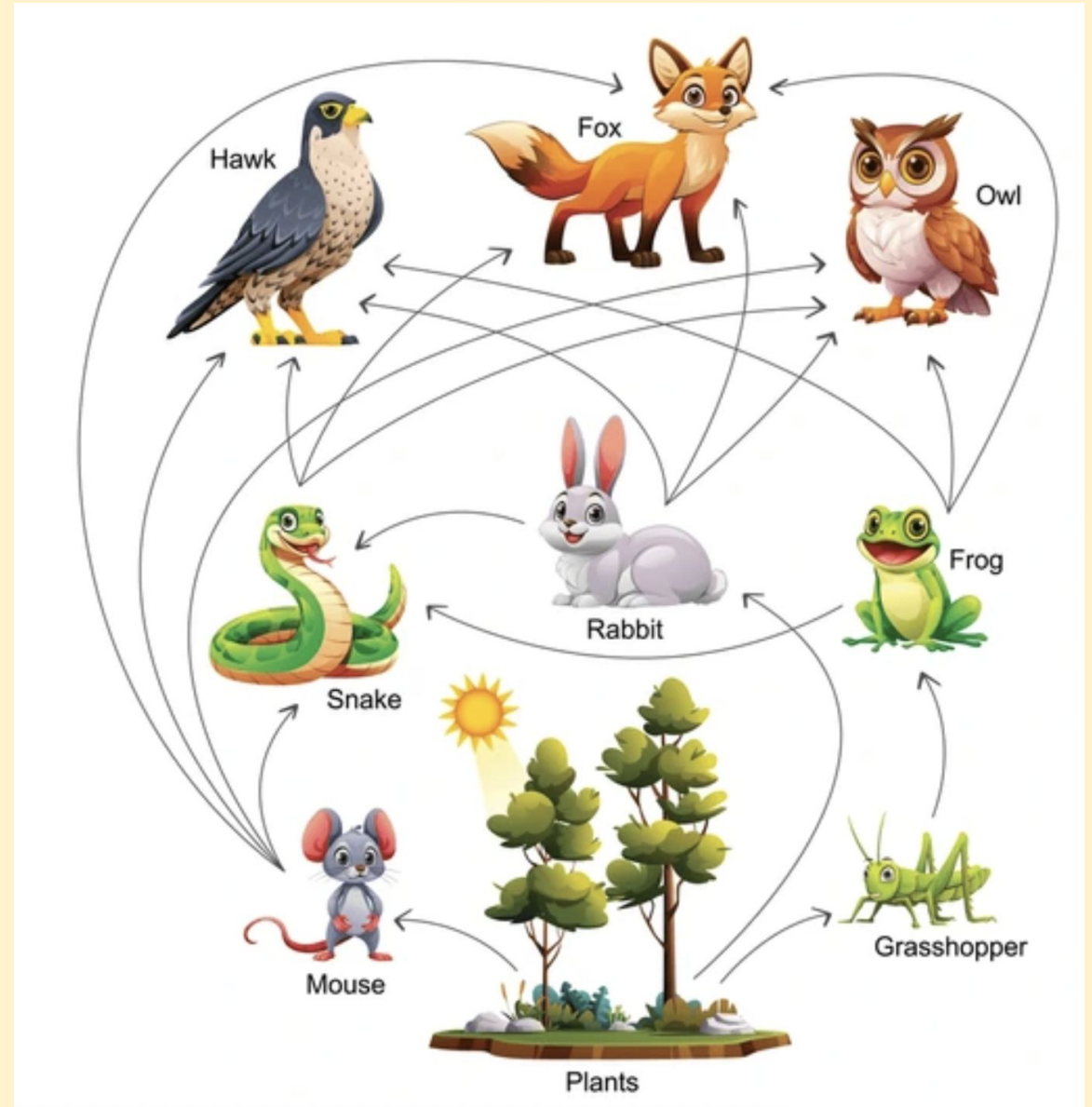
Food Chains

But food chains don't actually end with the top predators, think back to year 7, what is the missing link?



What is a food web?

A food web is a network of interconnecting food chains showing the energy flow through part of an ecosystem



Food webs

What would the food web look like for these food chains?

In pairs: Try to draw out the web, putting the plants at the bottom and working up

plants → aphid → ladybird → blue tit → owl

plants → moth larva → blue tit → owl

plants → moth larva → spider → chiffchaff → owl

plants → vole → stoat

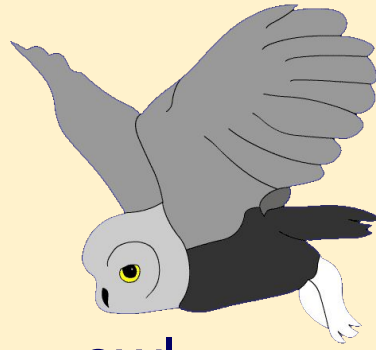
plants → vole → owl



10:00



blue tit



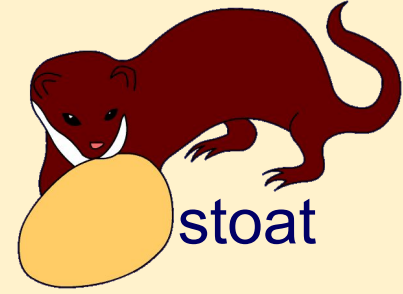
owl



chiffchaff



spider



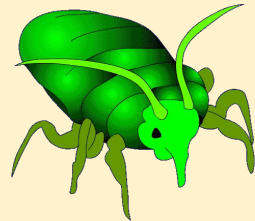
stoat



ladybird



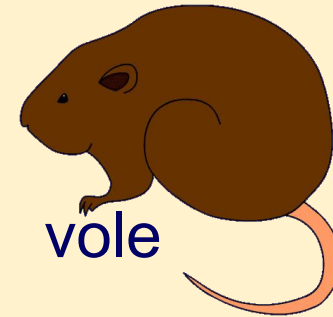
moth larva



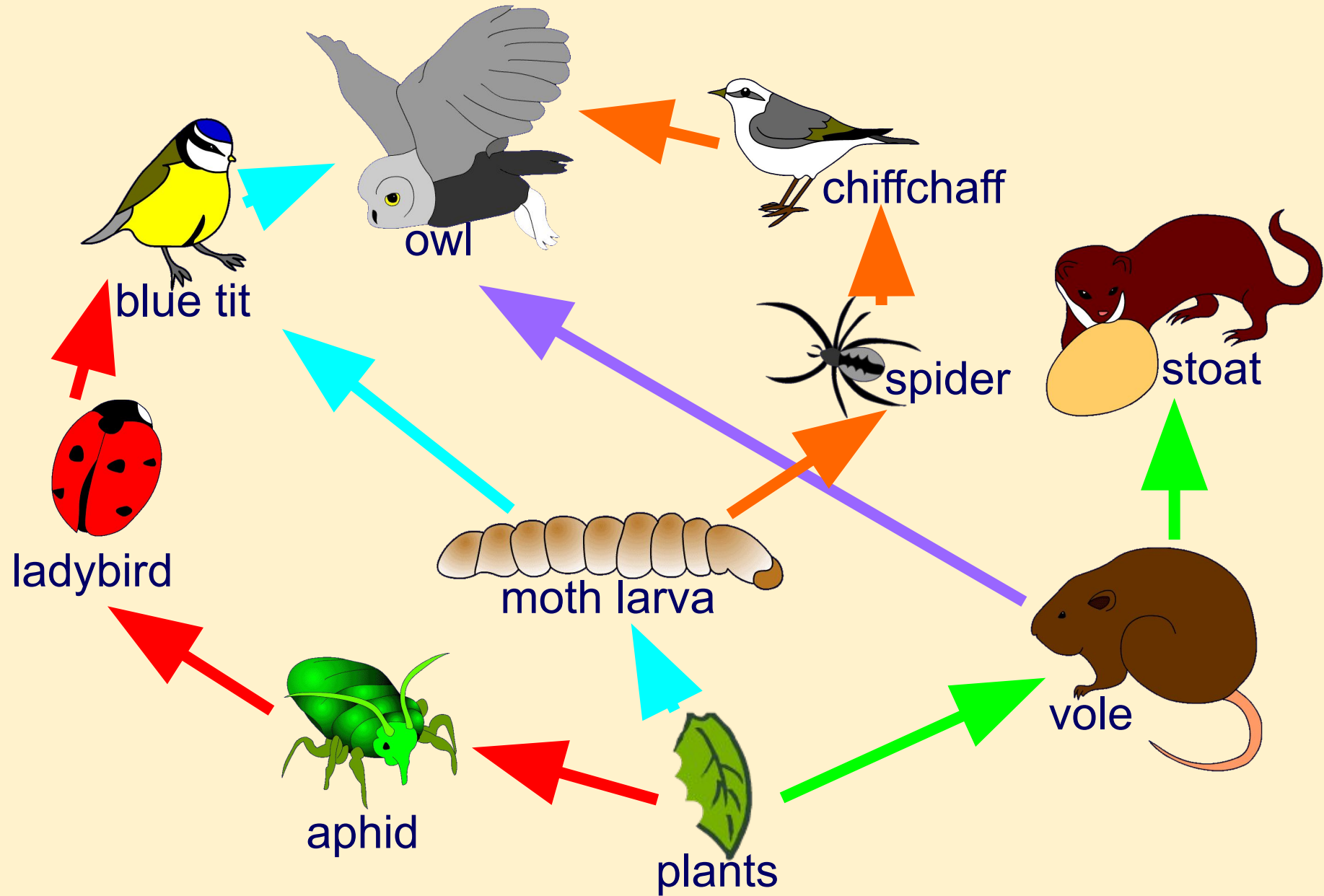
aphid



plants

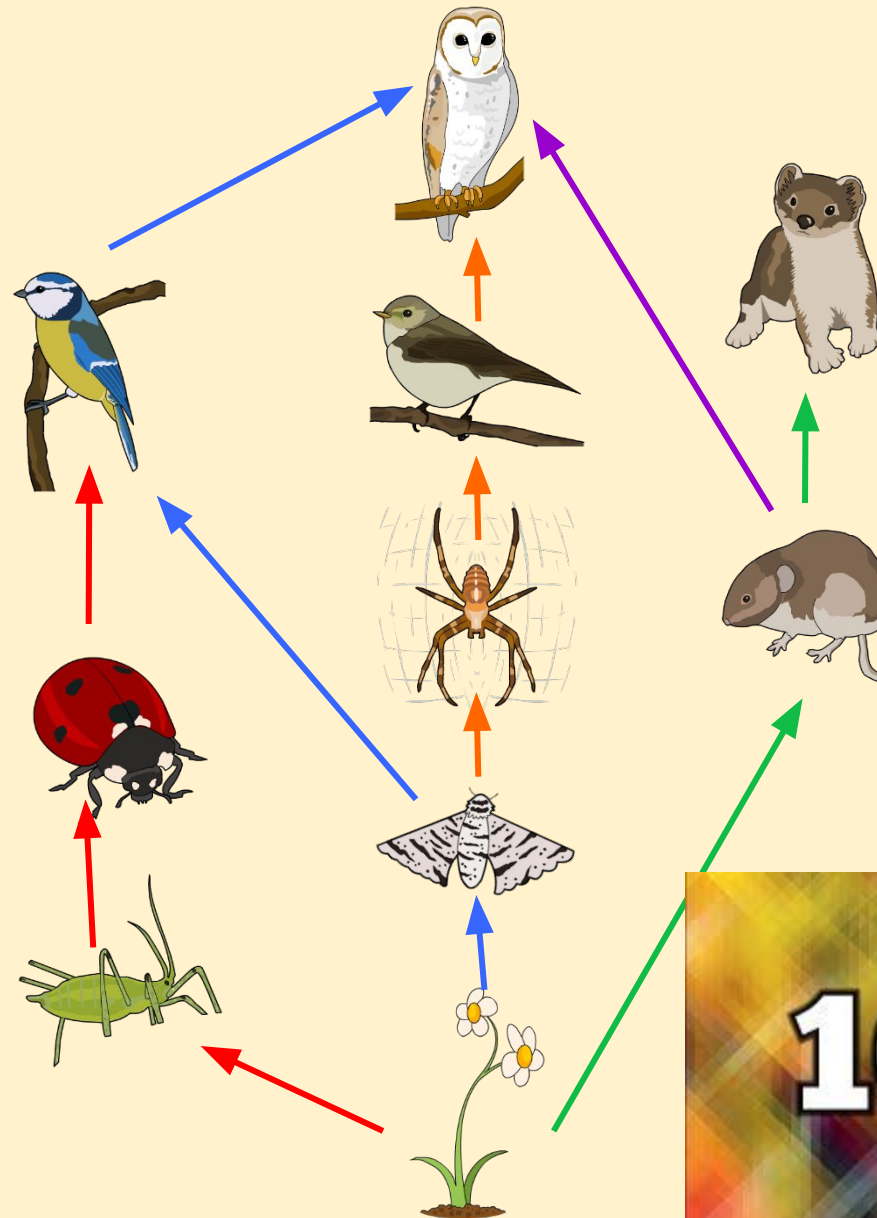


vole



Food webs

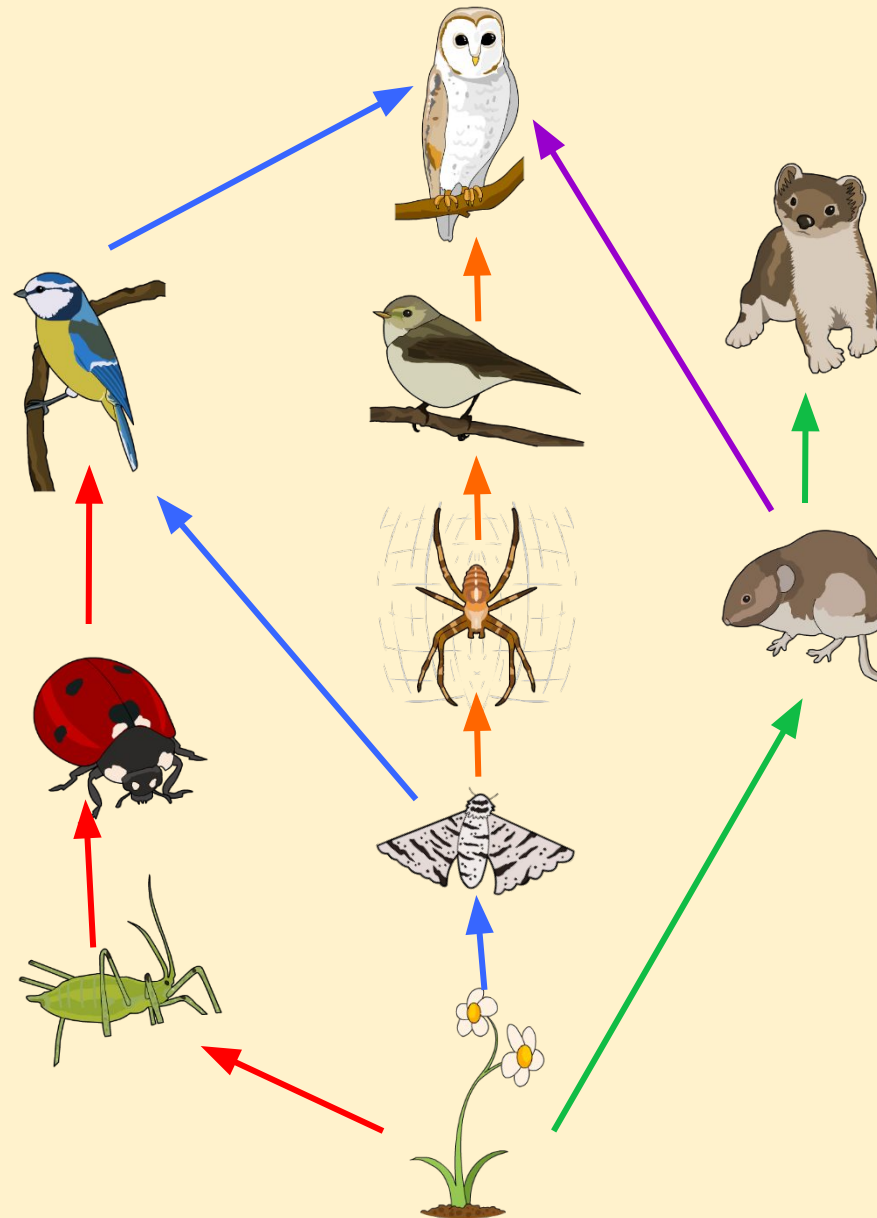
1. Name the producer in this food web.
2. Name two herbivores in this food web.
3. Name two species that are tertiary consumers
4. How many secondary consumers are there?
5. What would happen if most blue tits disappeared because of a disease?



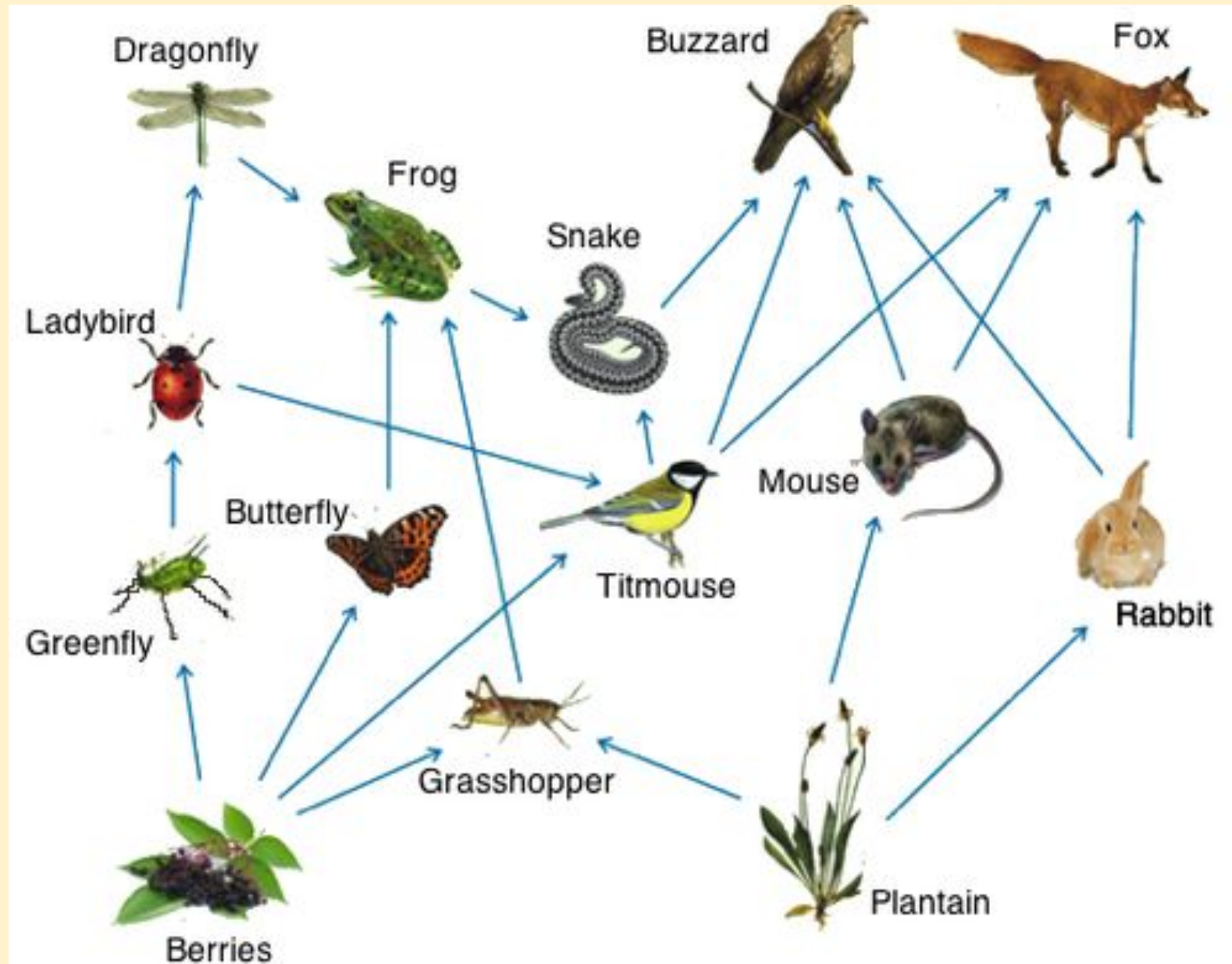
10:00

Food webs

1. Name the producer in this food web. **Daisy(flower/plant)**
2. Name two herbivores in this food web. **Vole and Aphid**
3. Name two species that are tertiary consumers **Stoat and barn owl**
4. How many secondary consumers are there? **Four (ladybird, spider, stoat, blue tit)**
5. What would happen if most blue tits disappeared because of a disease? **Ladybug and moth will increase**



What is the longest food chain in this food web?

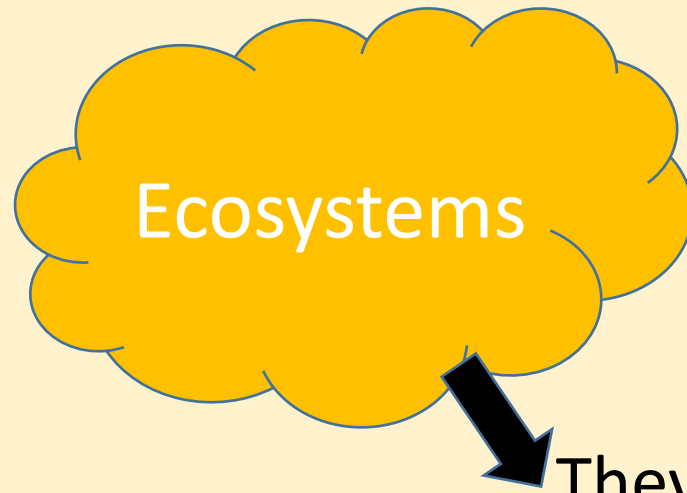


Ecosystems

Plenary: Go back to your mind map and add on in a different coloured pen, what you have learnt today.

Test yourself!

[Energy Flow in Ecosystems](#)
[Flashcards | Quizlet](#)



They can be little as a pond or as big as a rainforest.